Notice, question, predict

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use observations to make a prediction that can be tested.

SUCCESS CRITERIA - I can...

1. Separate stated evidence from unsupported claims.
2. Extend a position pattern and state an assumption.
3. Name an observation that could test the prediction.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “evidence” and “prediction” to explain your model or result. Sentence starters are optional.

evidence: observations or measurements that support an idea

prediction: what a model says may happen next

model: a drawing, explanation or equation for a physical situation

01 Read a scale

A ruler shows a toy car at 2 cm, then at 7 cm.
Draw both positions and the change. How far apart are the marks?

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta x = x_f - x_i$; $x_{next} = x_{now} + \Delta x$.
For a repeating position pattern only. i = initial; f = final. Check the pattern against evidence.

Worked reasoning: A cart is at 2, 5 and 8 m at 0, 1 and 2 s. Equal jumps of 3 m suggest 11 m at 3 s, if the pattern continues.

Your turn: A motion sensor recorded the first three positions in the table. Add the predicted position at 3 s.
Draw the four positions on a line and distinguish the prediction from the measurements.

Sentence starter: I predict ... . My model supports this because ... .

Time (s)	Position (m)
0	2
1	5
2	8
3	Predict: ____

Marzano check - circle: ① ② ③

03 What does the sensor show?

A class uses a motion sensor to track a cart. The table shows its position once each second.
Describe two patterns and circle the measurements that support each one.

Support: I notice ____. The data show ____.

Sentence starter: I notice ... . This supports ... because ... .

Time (s)	Position (m)
0	1
1	3
2	5

Marzano check - circle: ① ② ③

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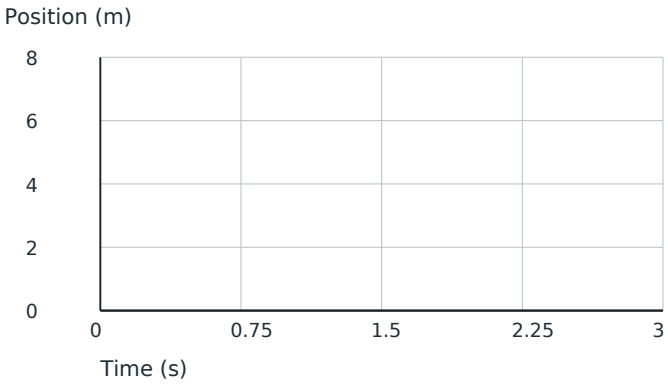
Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Make a prediction

Use the measurements in question 3 to predict where the cart will be at 3 s. Add the prediction to the blank graph. **Show** the pattern you used.

Sentence starter: I predict … . My model supports this because … .



Marzano check - circle: ① ② ③

05 Evidence or a claim?

A fictional space report says: "An object is approaching Earth, but its future path is uncertain." For each statement, write **supported** or **not established**:
(a) It is getting closer.
(b) It will definitely hit Earth.
(c) More observations are needed to predict its path.

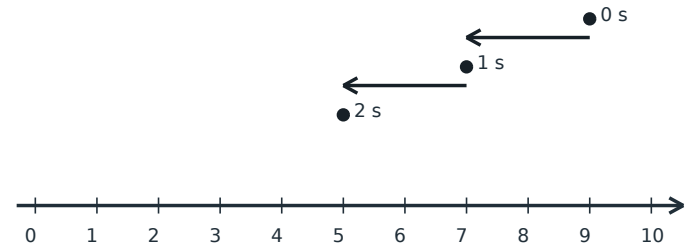
Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

06 Test the model

A different cart is observed at 9, 7, 5 m at 0, 1, 2 s. **Predict** its position at 3 s. What would a new reading of 4 m make you check?

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③

07 Ask a question you can investigate

A classmate asks, "Should we worry about the asteroid?" Rewrite this as a question that a measurement could help answer.
Name the measurement and explain how it would help.

Support: I could investigate ____. I would measure ____.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

08 Choose useful evidence

Two groups time the same toy car over the same marked track. One records three times; the other writes "pretty fast." Which record would let someone check the claim? **Explain**.

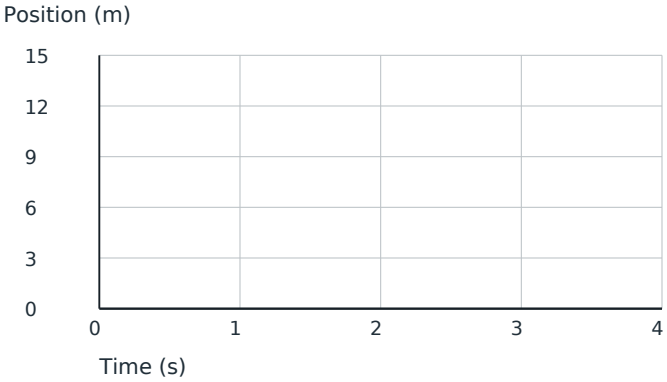
Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

09 Exit check

A robot moves along a marked floor. Its positions are 0, 3 and 6 m at 0, 1 and 2 s.
(a) Plot the measurements and predict its position at 4 s.
(b) State your assumption and one later observation that would challenge it.

Sentence starter: I predict … . My model supports this because … .

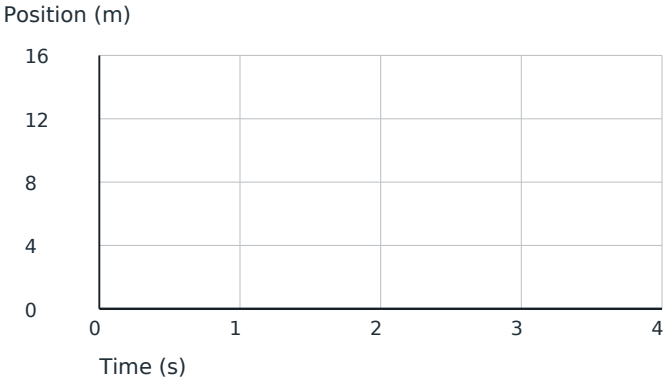


Marzano check - circle: 1 2 3

10 STRETCH: test the model

Return to the worked example in problem 2. Create two possible positions at 4 s if the cart changes its motion after 2 s.
Draw both possibilities. What measurement would help choose between them?

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: 1 2 3